Visualizing US Natural Disaster Declarations: Trends and Patterns

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1. Project Description

This Visualization of US Natural Disaster project focuses on analysing and visualizing natural disaster declarations across the United States using historical data from the Federal Emergency Management Agency (FEMA). The primary objective is to explore temporal trends, seasonal patterns, and incident-type distributions to better understand how disaster declarations have evolved over time.

The dataset contains detailed records of federally declared disasters, including information such as declaration number, state, incident type, incident begin and end dates, declaration date, and program assistance types. Using Python-based data analysis and visualization techniques, the project transforms raw disaster records into meaningful visual insights.

2. Dataset Used

The project uses the **FEMA Federal Disaster Declarations Dataset**, sourced from Kaggle. It contains historical records of disaster declarations issued by the Federal Emergency Management Agency (FEMA) across the United States.

This dataset provides structured information about federally declared disasters, including details about the type of incident, affected state, declaration dates, and assistance programs.

3. Environment Setup

The project is implemented in **Python** using **VS Code**. The key libraries required include:

- **Pandas –** For efficient numerical operations and tabular data handling.
- **Matplotlib and Pyplot –** Used for visualizing images, density maps, and data insights during exploration.

A local virtual environment was created to maintain package consistency and avoid conflicts.

4. Data Exploration

Before preprocessing and analysis, the disaster dataset was thoroughly explored to understand its structure, verify data integrity, and ensure suitability for time-based visualization. This step is crucial because incorrect or inconsistent data can lead to misleading analytical results.

4.1. Verifying dataset file and structure

- Checking whether the dataset exists in the file system
- Confirming correct file path for loading the CSV file
- Viewing the first few rows of the dataset
- Verifying available columns and data types.

4.2. Inspecting Sample Records

A sample of disaster declaration records was displayed to understand:

- Structure of each record.
- Presence of missing or inconsistent values.

4.3. Validating Time-Related columns

Since the project focuses on temporal trends, the date column was verified and converted into datetime format.

5. Data Preprocessing

This stage involves transforming raw disaster declaration records into analysis-ready data suitable for visualization and time-series analysis. The preprocessing pipeline includes the following key steps:

5.1. Dataset Loading

The dataset is loaded using pandas from the CSV file stored in the data/ directory.

Purpose:

- Ensures the dataset is successfully imported into a DataFrame.
- Allows inspection of structure and column types
- Prepares data for cleaning and transformation

5.2. Date Conversion and Validation

Since the project focuses on temporal analysis, date fields are converted into

proper datetime format.

Steps:

- Convert Date into datetime format.
- Handle invalid date entries using errors
- Check for missing date values

Purpose:

- Ensures accurate time-based analysis
- Prevents errors in trend and seasonality calculations

5.3. Feature Extraction

To analyze trends and seasonality, new time-related featured are extracted.

Purpose:

- Enables yearly and trend analysis
- Supports monthly seasonality visualization

5.4. Handling Missing and Inconsistent Data

The dataset is checked for missing values and inconsistencies.

Steps:

1. Identify null values in important columns
2. Remove or handle rows with missing critical data
3. Check for duplicate records

Purpose:

- Improves data reliability
- Prevents misleading visualizations
- Ensures clean input for aggregation and plotting

5.5. Data Aggregation for Visualization

For trend comparison and analysis, the dataset

is grouped and summarized.

Steps:

- Group disaster declarations by year
- Group by incident type for comparison

Purpose:

- Converts raw records into summarized insights
- Prepares structured data for visualizations
- Enables clear comparison across time disaster types

6. Proposed System

1. Raw High-Resolution Image

The system begins with the FEMA Federal Disaster Declarations dataset containing historical records of disaster events across the United States.
The dataset includes:
• Declaration Number
• State
• Incident Type
• Incident Begin Date
• Incident End Date
• Declaration Date
• Assistance Program Information
These raw records require cleaning and transformation before analysis.

2. Preprocessing (Cleaning & Feature Engineering)

The raw dataset is processed to ensure consistency and reliability.

This ensures structured and analysis-ready data for visualization.

3. Data Aggregation & Trend Analysis

The cleaned dataset is grouped and summarized to identify meaningful patterns.

Operations include:

- Year-wise disaster declaration count
- Month-wise aggregation for seasonality detection.
- Incident-type comparison across years
- State-wise disaster frequency analysis

7. Training Pipeline

Analysis Pipeline:

Features:

- Automated data loading and validation
- Date conversion and feature extraction
- Year-wise and month-wise aggregation
- Incident-type comparison
- Generation of trend and seasonal visualizations

Why Time-based training?

- Disaster analysis requires identifying patterns over time.
- Aggregating by year reveals long-term growth or decline
- Aggregating by month helps detect seasonal patterns
- Grouping by incident type enables comparative trend analysis

Why use Incident Begin Date?

- It represents the actual start of a disaster event.
- More reliable for seasonality analysis.
- Avoids bias that may occur from declaration date delays.

8. Conclusion

The *Visualization of US Natural Disaster Declarations* project successfully analysed historical FEMA disaster data to identify meaningful temporal patterns and trends. By transforming raw disaster records into structured, time-based insights, the project provided a clear understanding of how disaster declarations have evolved over the years.

Through systematic data exploration, preprocessing, and aggregation, key patterns such as yearly growth trends, seasonal variations, and incident-type distributions were identified. The analysis revealed noticeable fluctuations in disaster declarations across different time periods and highlighted the dominant disaster categories affecting the United States.

The visualization approach helped convert complex tabular data into intuitive graphical representations, making trends and comparisons easier to interpret. This project demonstrates how data analytics and visualization techniques can support better understanding of disaster frequency and seasonality.